SIMATS ENGINEERING
CO3- ASSESSMENT TOOL 2
SIMULATION TASK

COURSE NAME: DIGITAL SIGNAL PROCESSING COURSE CODE: ECA09

COURSE OUTCOME COVERED

CO 3: Evaluate finite word-length effects and multirate signal processing techniques, including decimation, interpolation, and polyphase decomposition. **Bloom Level mapped- BL3,BL4,BL5**

Topics Covered:
Quantization- Truncation and Rounding errors - Quantization noise - Limit cycle oscillations – Signal scaling.Parametric and Non parametric Spectral estimation. Decimation-Interpolation-Sampling rate conversion-Implementation of sampling rate conversion-Polyohase Decomposition -5G Decimation/Interpolation: For mmWave beamforming.

Assessment Tool number	**: CO3-AT2**
Assessment Tool Weightage	**: 35%**
Assessment Tool Name	**: Simulation Task**
Duration of this assessment	**: 60 Minutes**
Marks	**: 25 Marks**
Type of Assessment Conduction	**: Self Learning (Home Task)**

- Each student will be assigned one simulation-based problem related to the course topics.
- Solve the assigned simulation problem independently.
- Develop and run the simulation using MATLAB/Simulink.
- Analyze and present the results with graphs and screenshots.
- Upload the simulation code and output to GitHub.
- Submit the GitHub repository link in Google Classroom (GCR).

Dr.

Code of Conduct:

I _____________________(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Simulation tools					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	20%	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts
Model Design	25%	Develops accurate, well-structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools
Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results
Documentation	15%	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation

QUESTION :

23. Combined Decimation and Interpolation
Simulate a multirate system using decimation and interpolation and analyze the resulting sampling rate conversion. (25) (BL4) (WK4)
Given Data:

- Decimation Factor = 3
- Interpolation Factor = 2
- Input Sampling Rate = 18 kHz

Tasks:

1. Design an anti-alias filter.
2. Perform decimation.
3. Implement interpolation.
4. Analyze the output spectrum.
5. Determine the final sampling rate.
6. Compare input and output signals.

Solution:

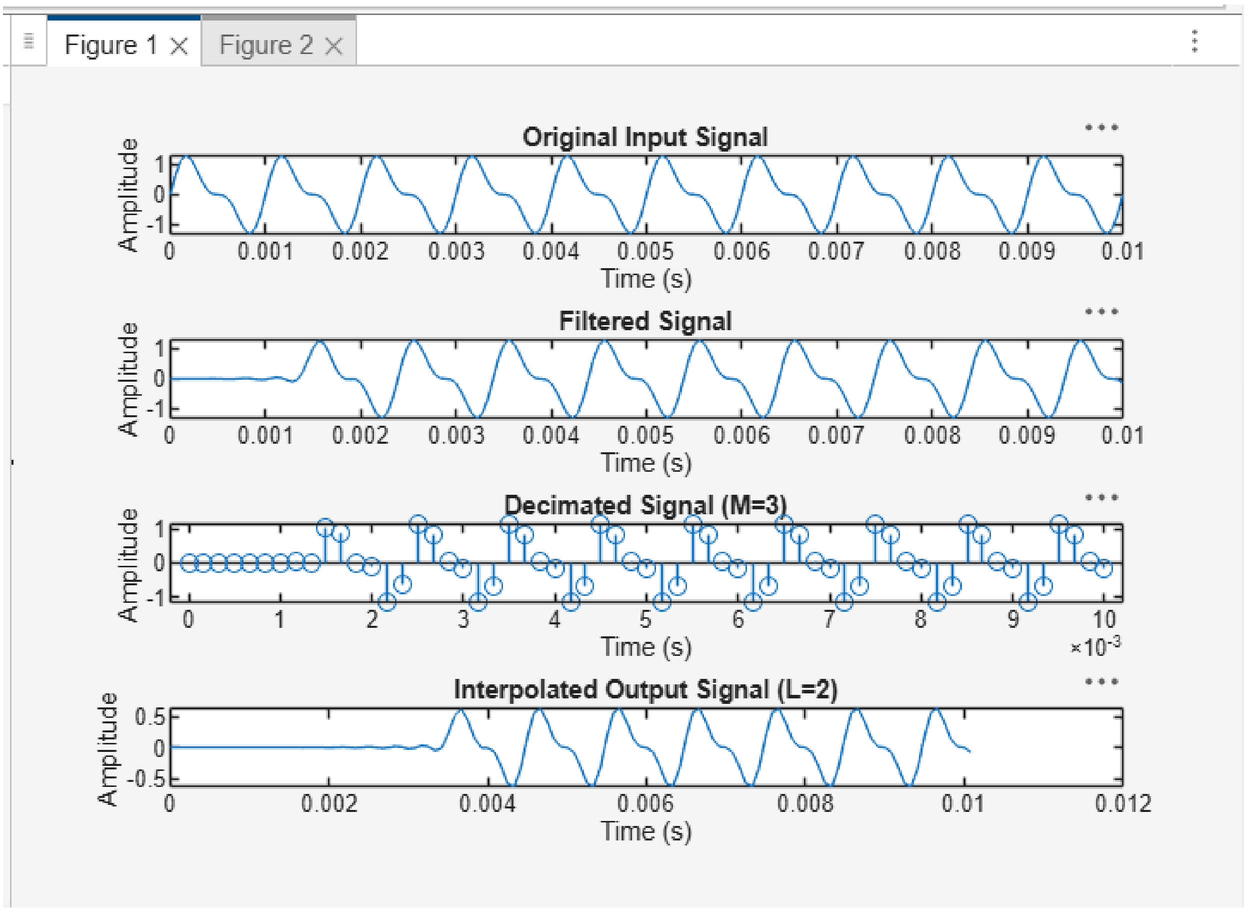
PROGRAMING CODE (MATLAB)

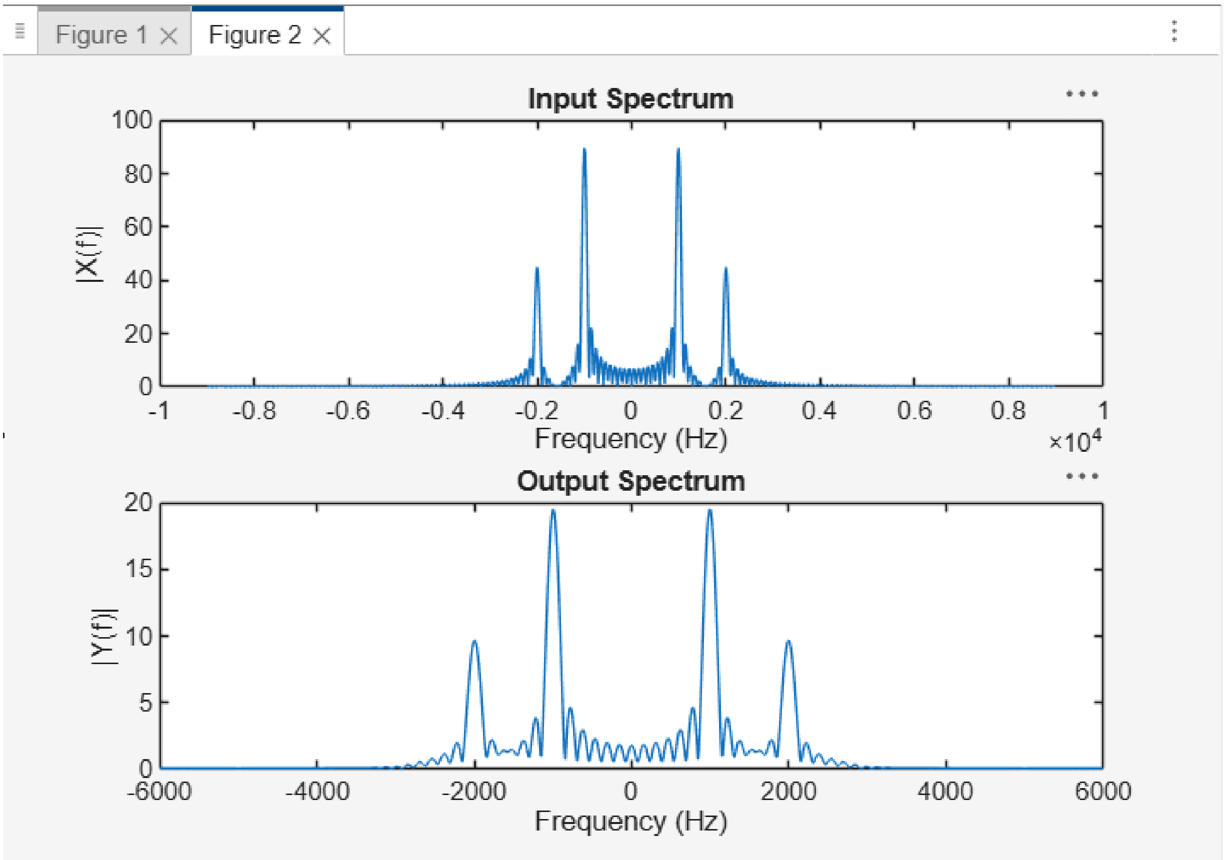
```
% Combined Decimation and Interpolation
% Fs = 18 kHz, M = 3, L = 2

clc;
clear;
close all;

%% Input Signal
Fs = 18000;          % Input Sampling Rate
t = 0:1/Fs:0.01;     % 10 ms duration

% Signal containing frequencies below cutoff
x = sin(2*pi*1000*t) + 0.5*sin(2*pi*2000*t);

%% Anti-Alias Low Pass Filter
fc = 3000;                     % Cutoff Frequency
Wn = fc/(Fs/2);                  % Normalized Cutoff
N = 50;                        % Filter Order

b = fir1(N,Wn,'low');              % FIR LPF
x_filt = filter(b,1,x);

%% Decimation (M = 3)
M = 3;
x_dec = downsample(x_filt,M);
Fs_dec = Fs/M;
```

```
%% Interpolation (L = 2)
L = 2;
x_int = upsample(x_dec,L);

% Reconstruction Filter
Fs_out = Fs_dec*L;
fc2 = Fs_dec/2;

Wn2 = fc2/(Fs_out/2);
b2 = fir1(N,Wn2,'low');

x_out = filter(b2,1,x_int);

%% Final Sampling Rate
fprintf('Input Sampling Rate = %d Hz\n',Fs);
fprintf('After Decimation = %d Hz\n',Fs_dec);
fprintf('Final Output Sampling Rate = %d Hz\n',Fs_out);

%% Time Domain Plots
figure;

subplot(4,1,1);
plot(t,x);
title('Original Input Signal');
xlabel('Time (s)');
ylabel('Amplitude');

subplot(4,1,2);
plot((0:length(x_filt)-1)/Fs,x_filt);
title('Filtered Signal');
xlabel('Time (s)');
ylabel('Amplitude');

subplot(4,1,3);
stem((0:length(x_dec)-1)/Fs_dec,x_dec);
title('Decimated Signal (M=3)');
xlabel('Time (s)');
ylabel('Amplitude');

subplot(4,1,4);
plot((0:length(x_out)-1)/Fs_out,x_out);
title('Interpolated Output Signal (L=2)');
xlabel('Time (s)');
ylabel('Amplitude');

%% Spectrum Analysis
figure;

subplot(2,1,1);
Nfft = 1024;
f1 = Fs*(-Nfft/2:Nfft/2-1)/Nfft;
X = fftshift(abs(fft(x,Nfft)));
```

```
plot(f1,X);
title('Input Spectrum');
xlabel('Frequency (Hz)');
ylabel('|X(f)|');

subplot(2,1,2);
f2 = Fs_out*(-Nfft/2:Nfft/2-1)/Nfft;
Y = fftshift(abs(fft(x_out,Nfft)));
plot(f2,Y);
title('Output Spectrum');
xlabel('Frequency (Hz)');
ylabel('|Y(f)|');
```

OUTPUT:





Command Window

```
Input Sampling Rate = 18000 Hz
After Decimation = 6000 Hz
Final Output Sampling Rate = 12000 Hz
>>
```